Building Multi-Tenant SaaS Architecture with Amazon EKS Capabilities and Auto Mode

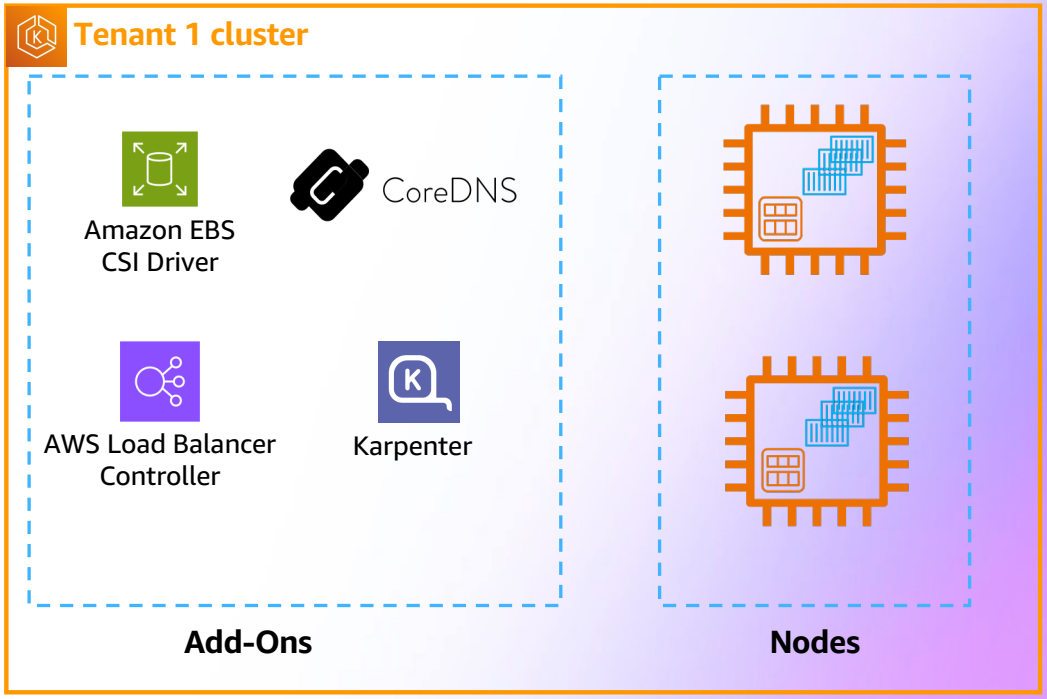
Chakkree Tipsupa

Solutions Architect
AWS

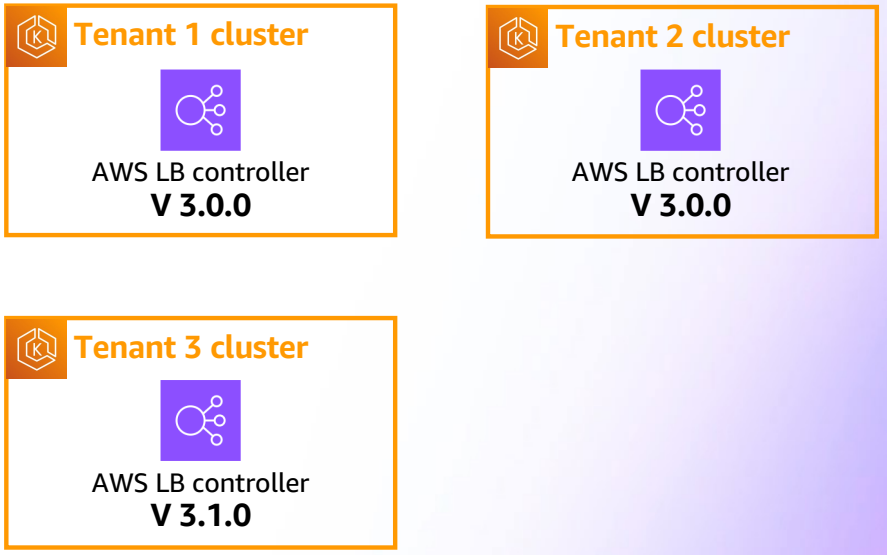
Rodrigo Bersa

Sr. Specialist SA, Containers
AWS

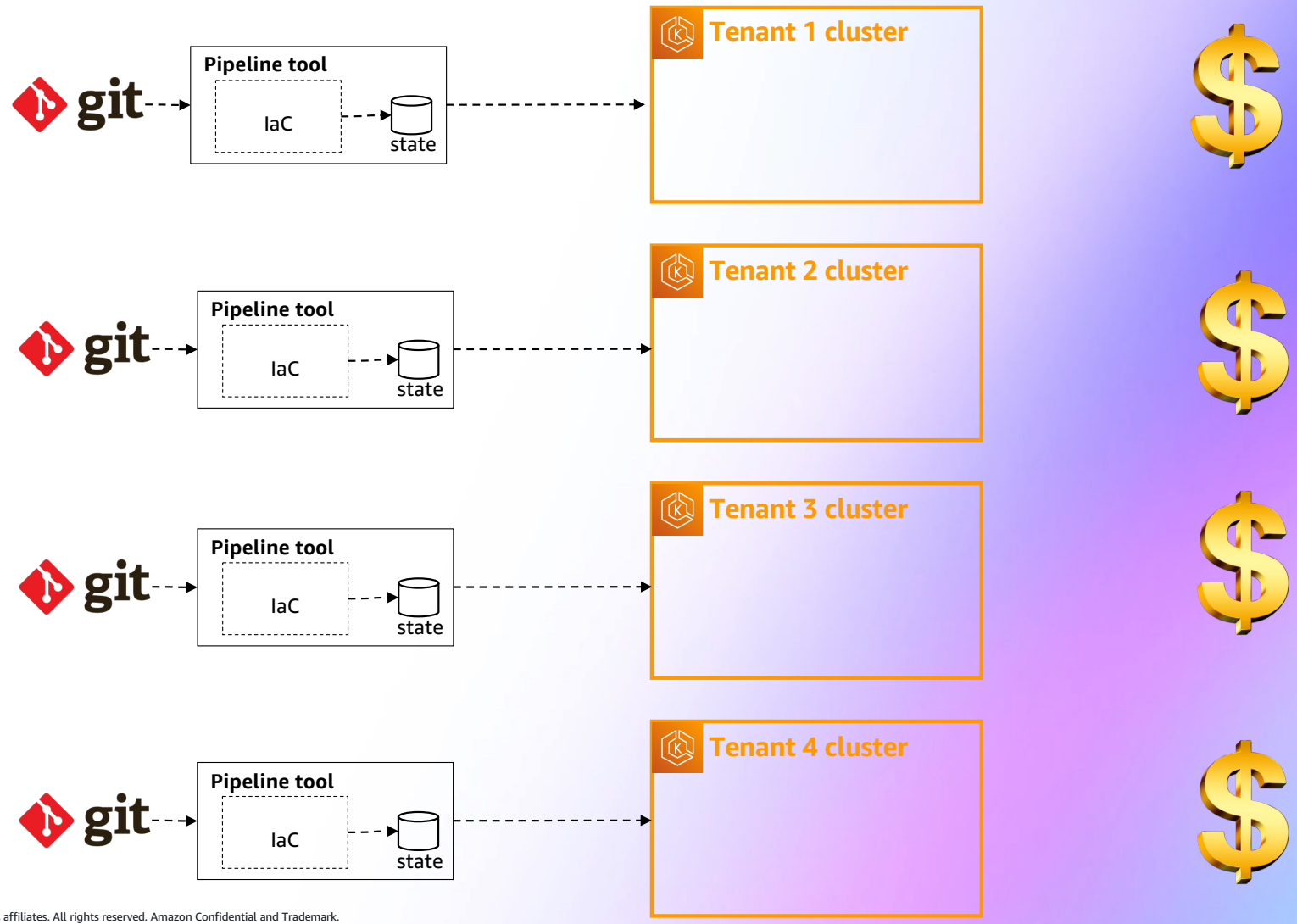
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AMAZON EKS

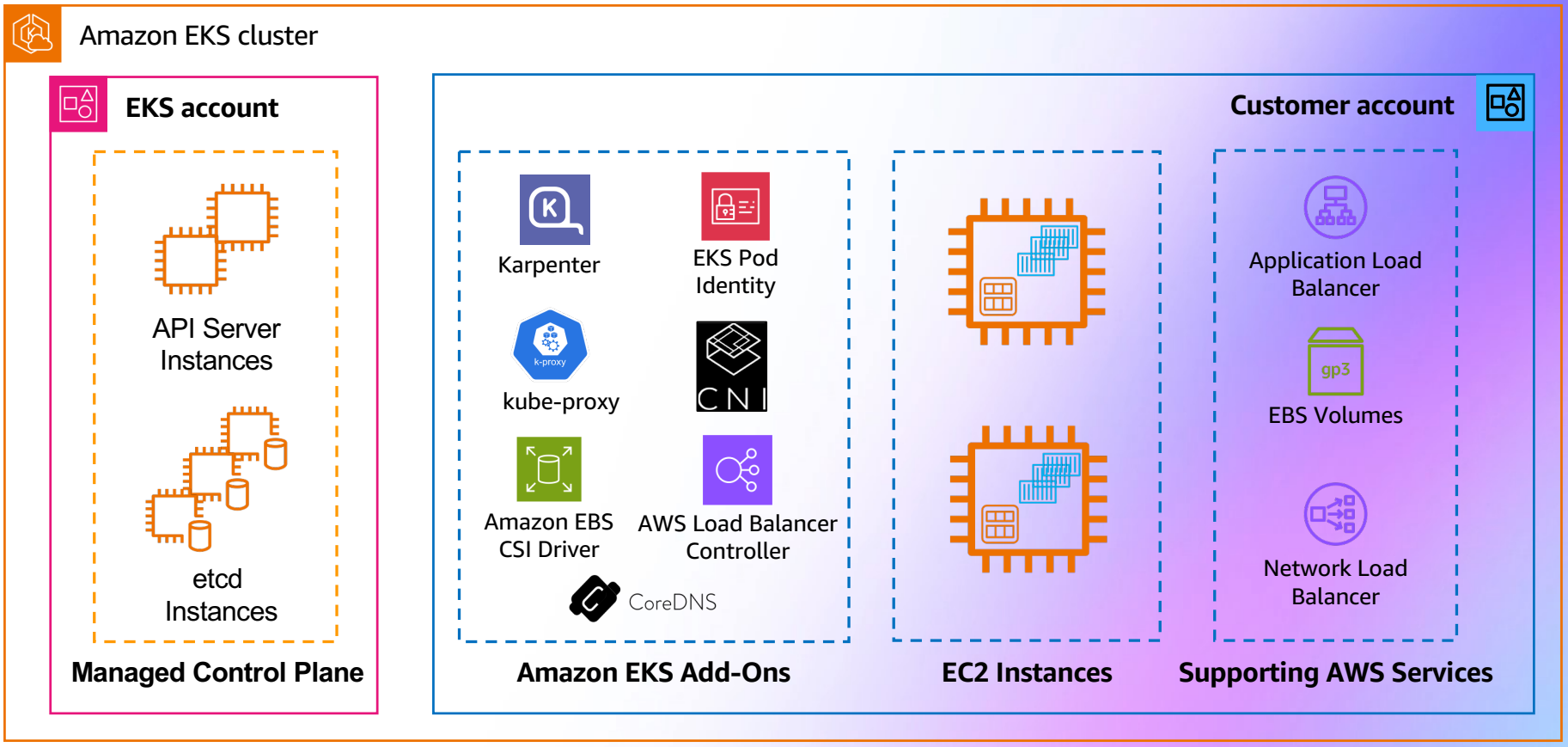
Auto Mode

Automate your entire Kubernetes cluster infrastructure

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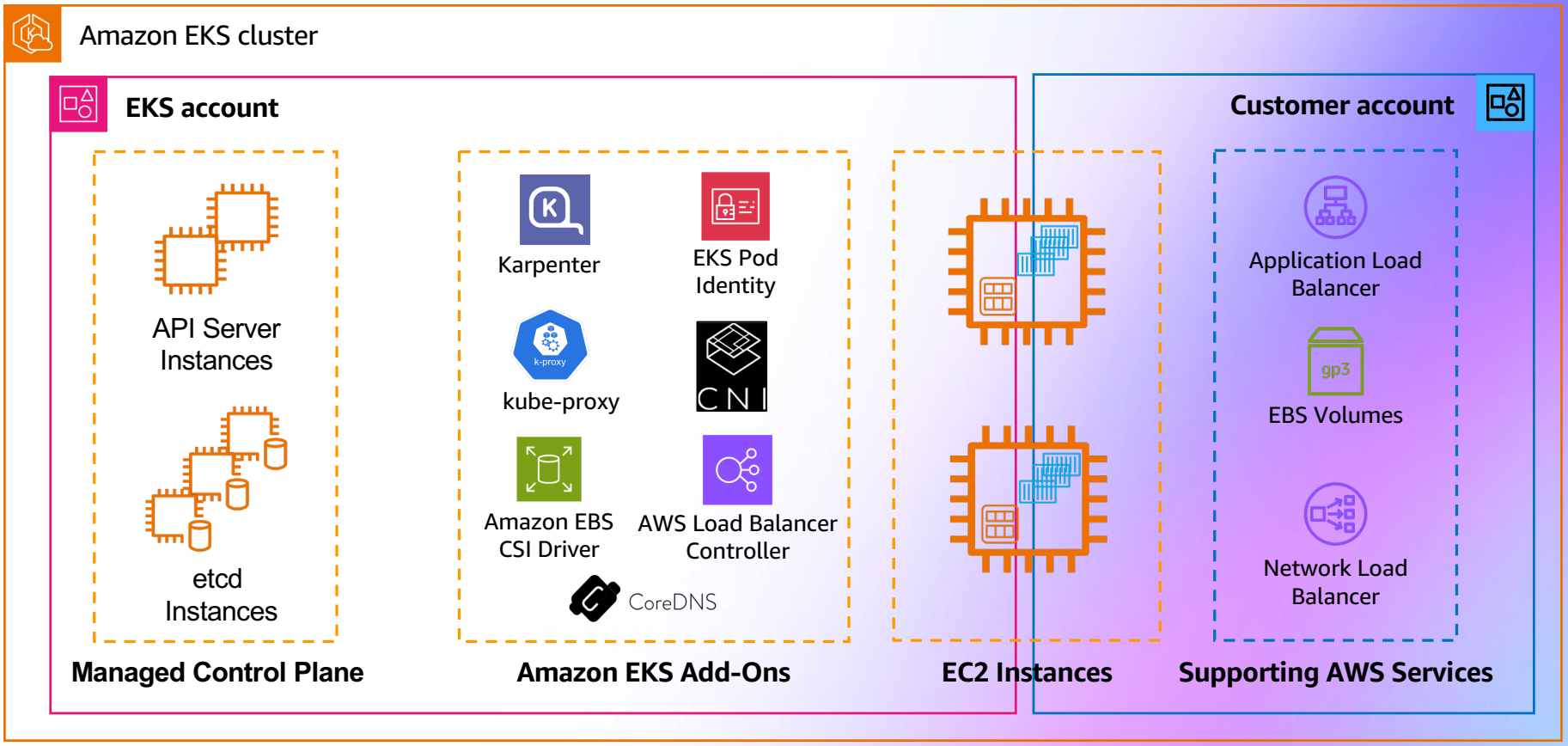
aws

Amazon EKS cluster architecture previously



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Amazon EKS cluster architecture with Auto Mode



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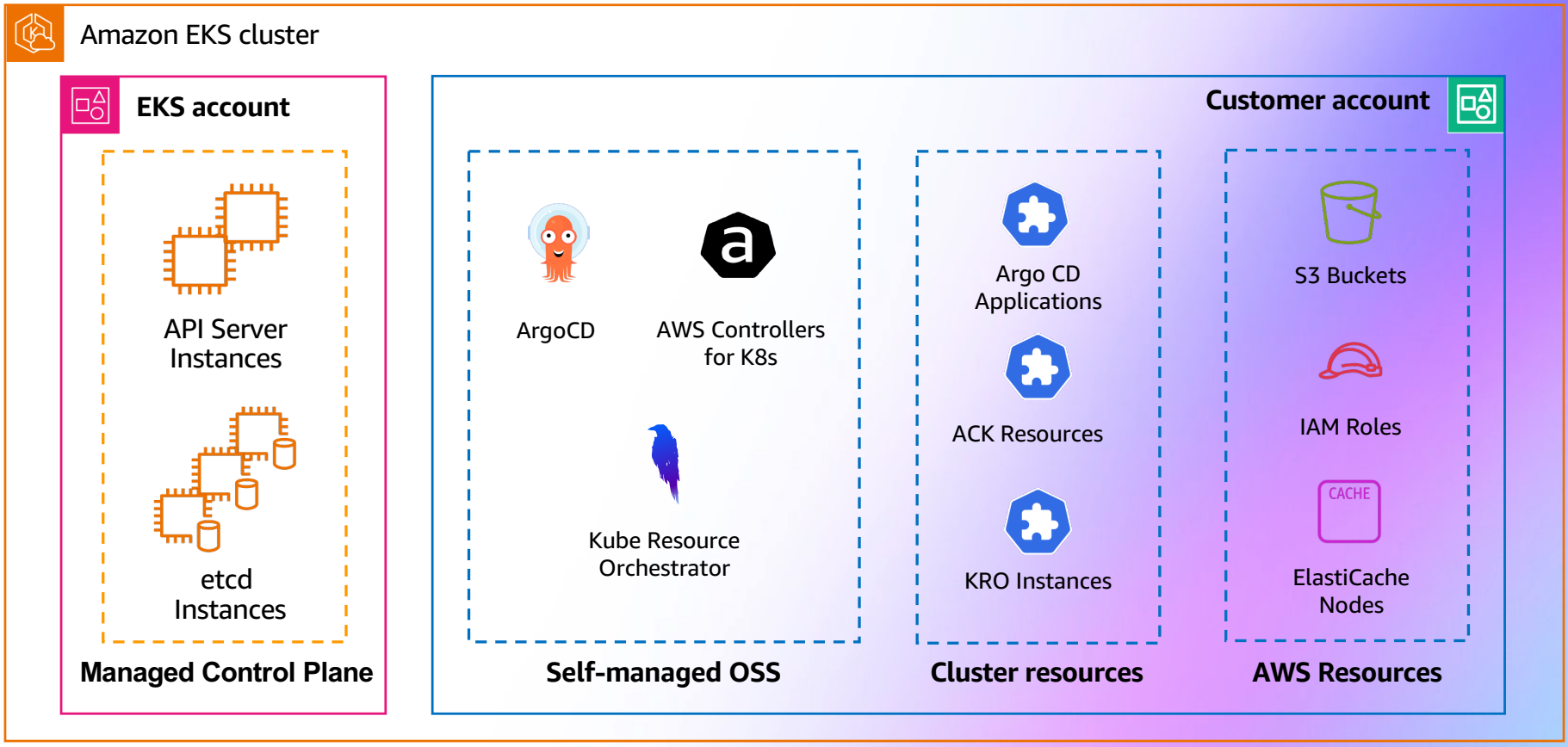
AMAZON EKS

Capabilities

Build and scale with Kubernetes

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Self-managed platforms with Amazon EKS



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Amazon EKS Capabilities

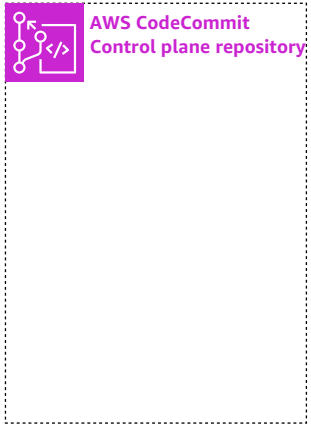


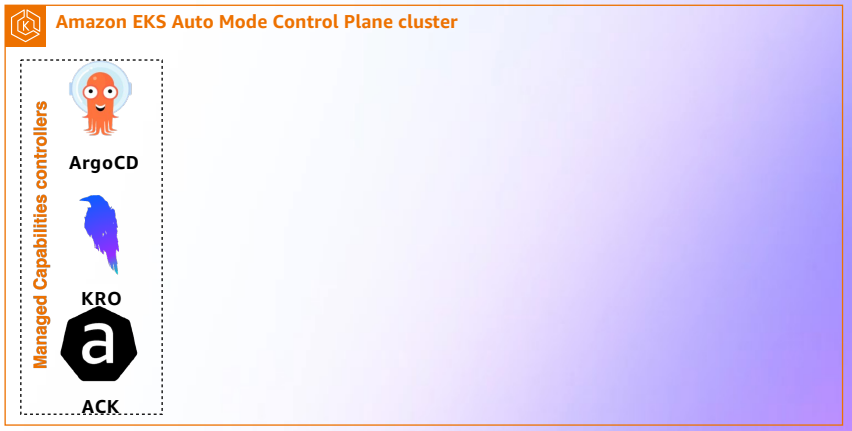
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How we put them together to solve other challenges?

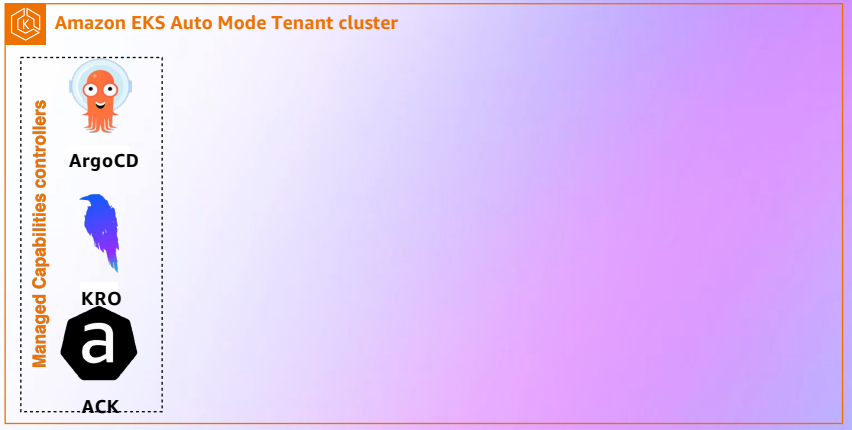
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Control plane and tenant architecture



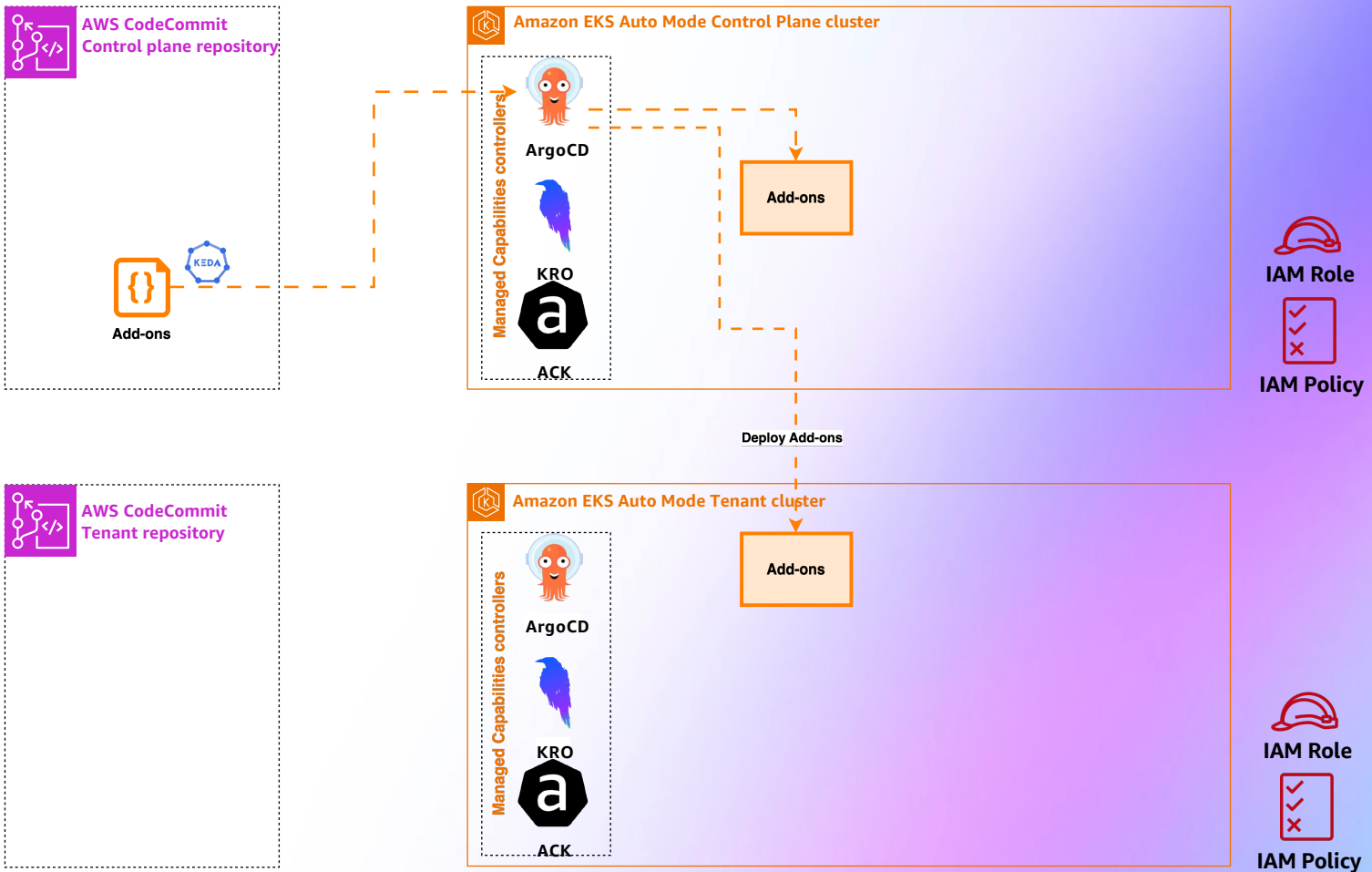




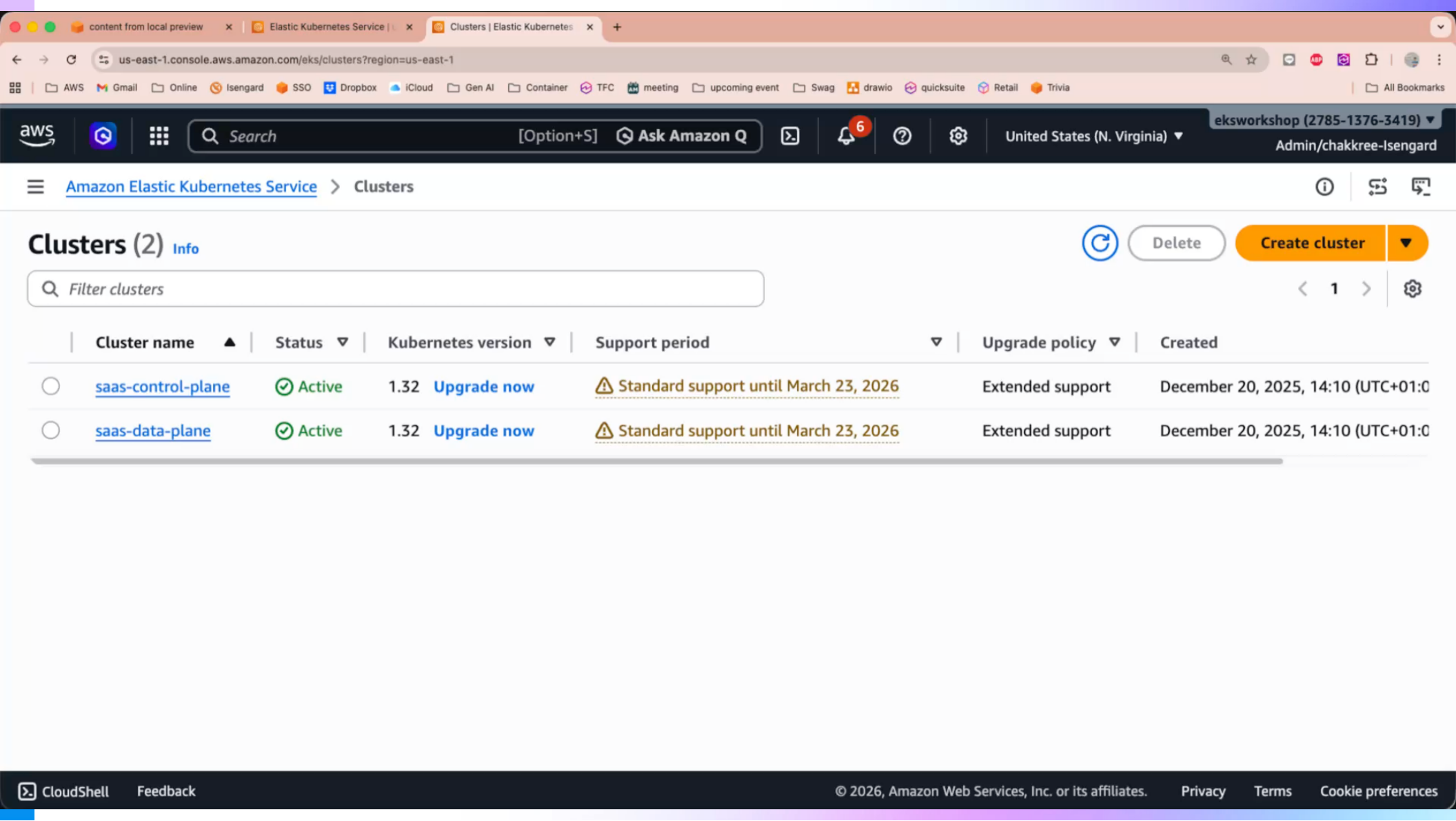


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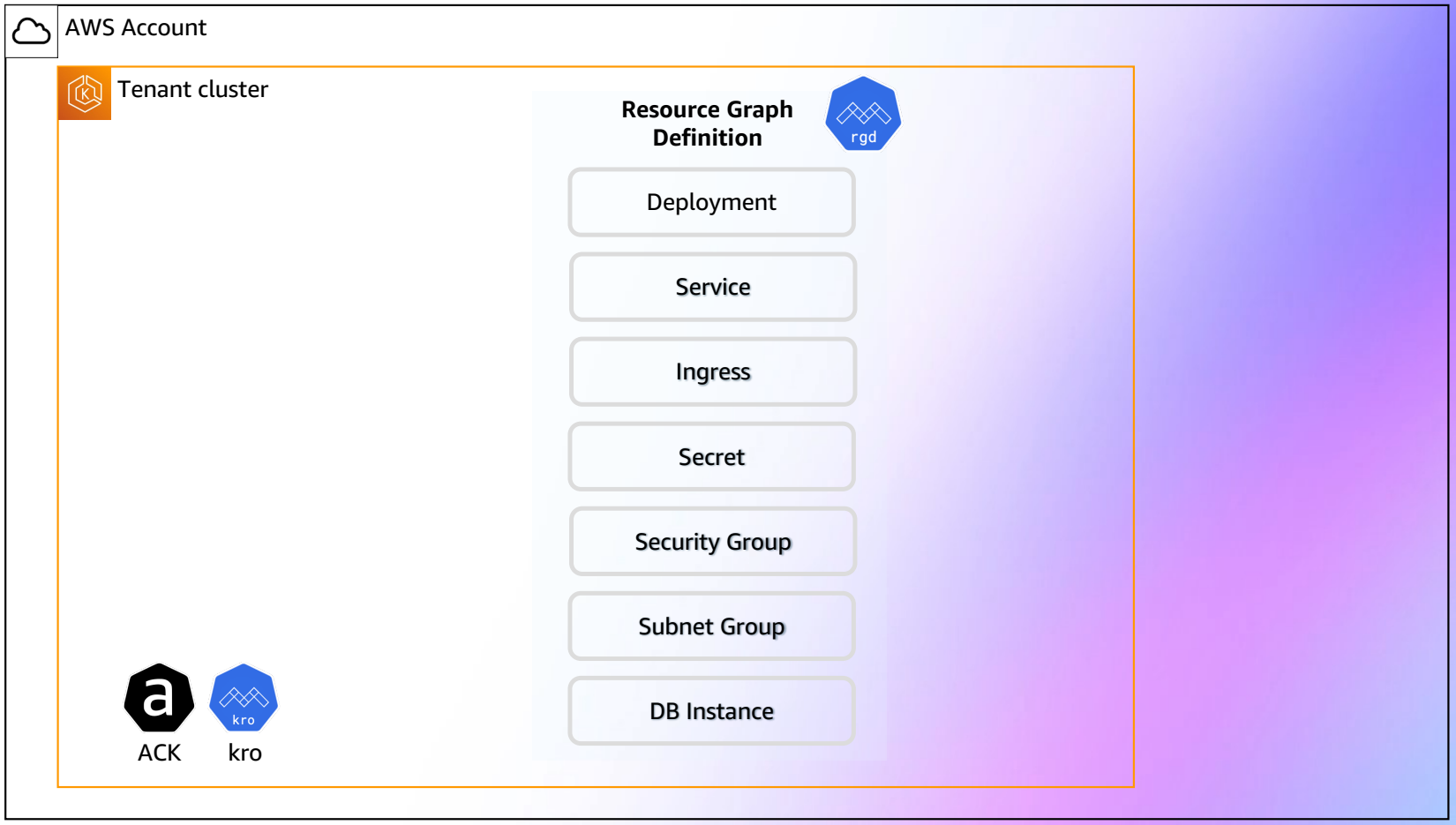
Deploying add-ons



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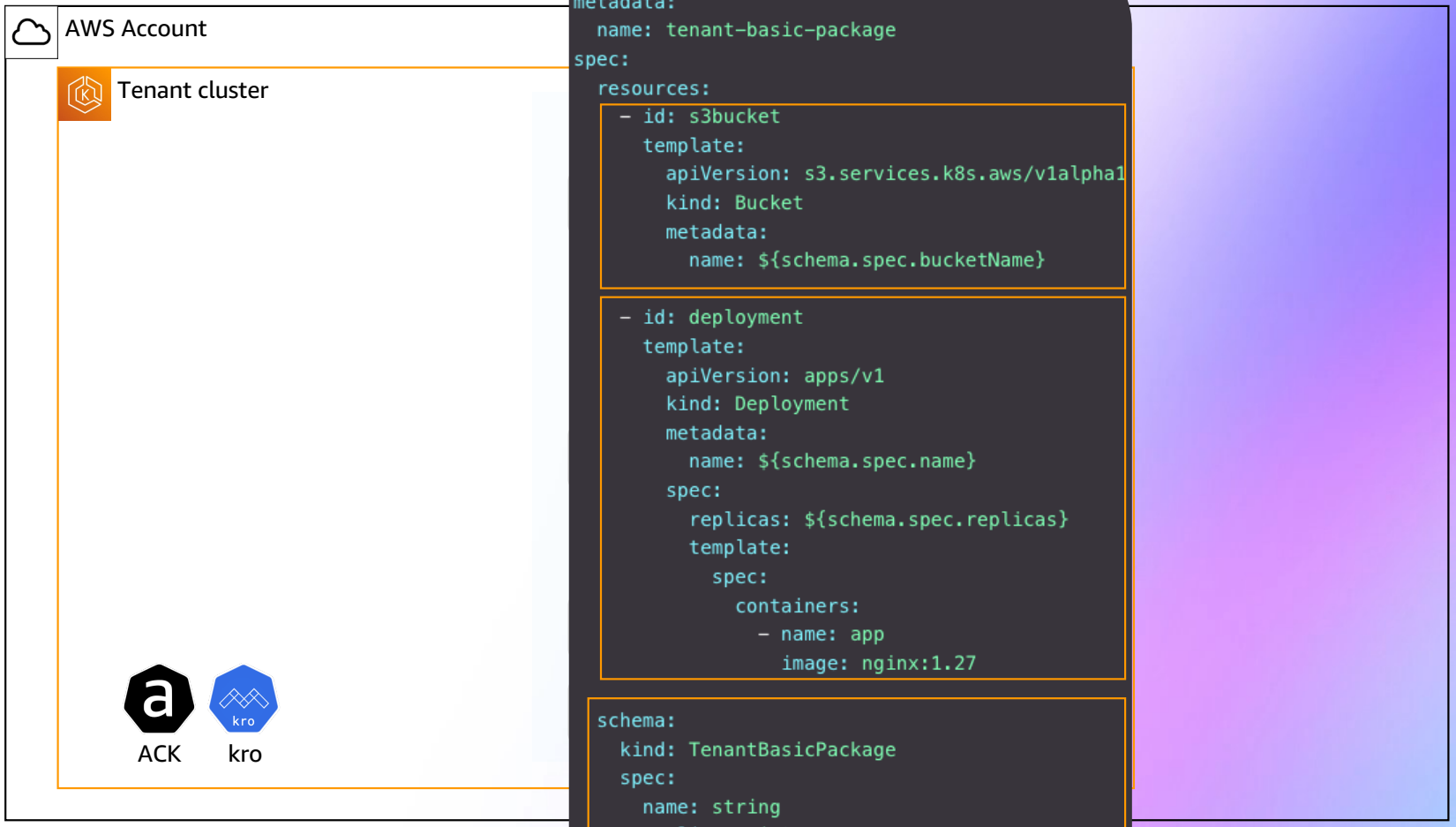

content from local preview
Elastic Kubernetes Service
Clusters | Elastic Kubernetes
us-east-1.console.aws.amazon.com/eks/clusters?region=us-east-1
AWS
Gmail
Online
Isengard
SSO
Dropbox
iCloud
Gen AI
Container
TFC
meeting
upcoming event
Swag
drawio
quicksuite
Retail
Trivia
All Bookmarks
aws
Search
[Option+S]
Ask Amazon Q
6
United States (N. Virginia)
eksworkshop (2785-1376-3419)
Admin/chakkree-Isengard
Amazon Elastic Kubernetes Service
Clusters
Clusters (2) Info
Delete
Create cluster
Filter clusters
1
Cluster name
Status
Kubernetes version
Support period
Upgrade policy
Created
saas-control-plane
Active
1.32
Upgrade now
Standard support until March 23, 2026
Extended support
December 20, 2025, 14:10 (UTC+01:0
saas-data-plane
Active
1.32
Upgrade now
Standard support until March 23, 2026
Extended support
December 20, 2025, 14:10 (UTC+01:0
CloudShell
Feedback
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How ACK and KRO work in a nutshell



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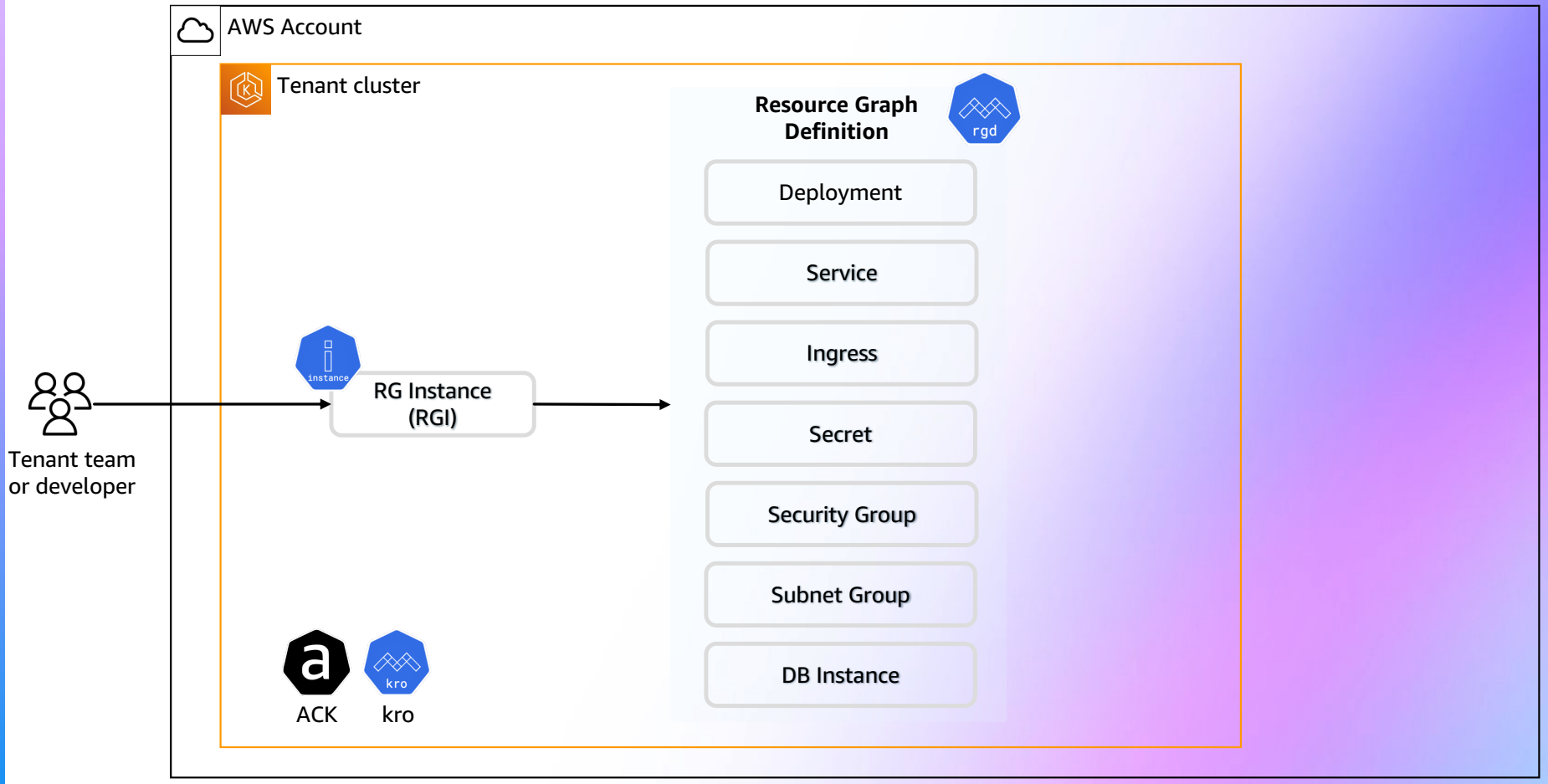
How ACK and KRO work in a nutshell




```yaml
apiVersion: kro.run/v1alpha1
kind: ResourceGraphDefinition
metadata:
  name: tenant-basic-package
spec:
  resources:
   - id: s3bucket
     template:
       apiVersion: s3.services.k8s.aws/v1alpha1
       kind: Bucket
       metadata:
         name: ${schema.spec.bucketName}
   - id: deployment
     template:
       apiVersion: apps/v1
       kind: Deployment
       metadata:
         name: ${schema.spec.name}
       spec:
         replicas: ${schema.spec.replicas}
         template:
           spec:
             containers:
               - name: app
                 image: nginx:1.27
  schema:
    kind: TenantBasicPackage
    spec:
      name: string
      replicas: integer
      bucketName: string
```

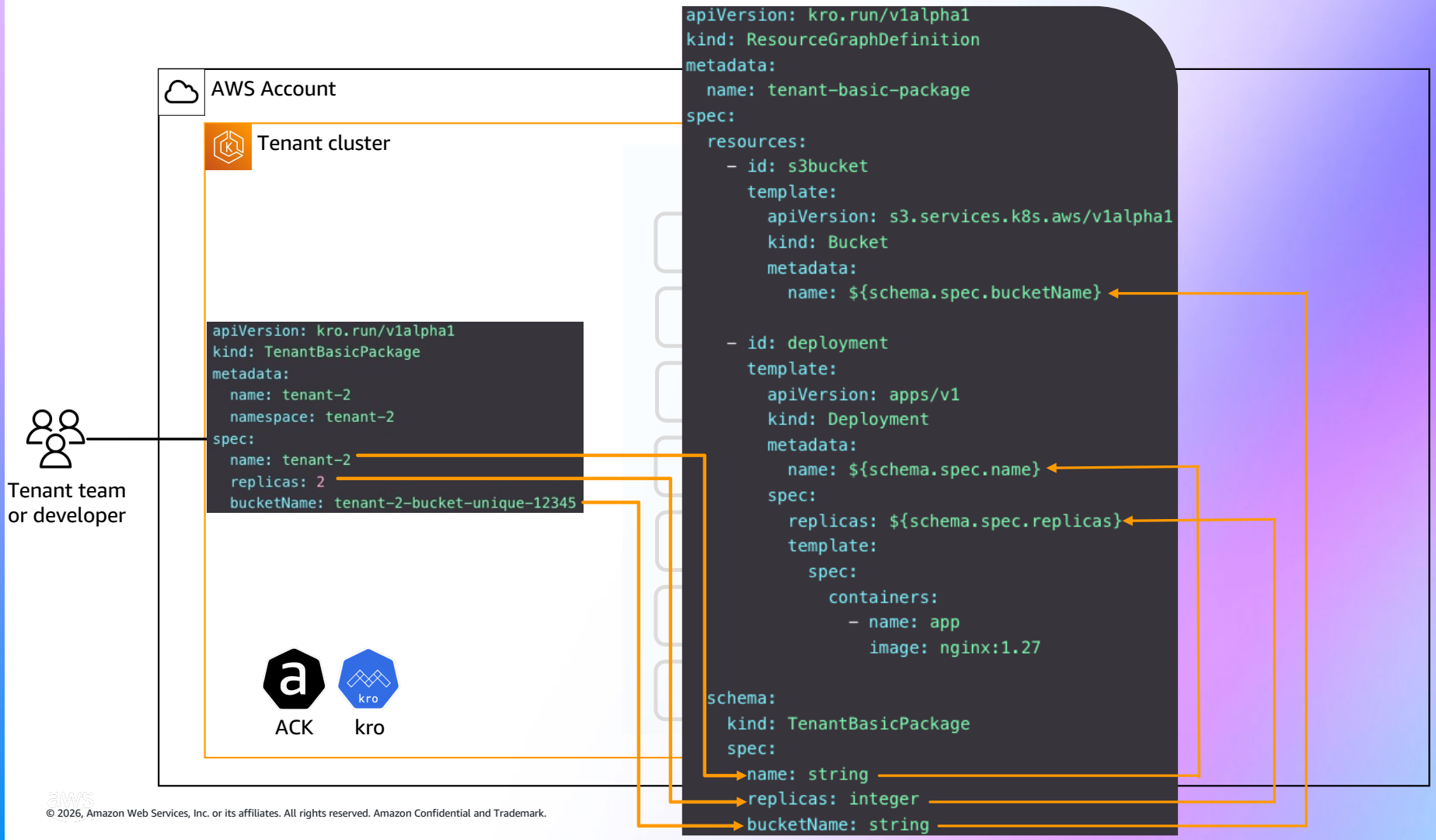
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How ACK and KRO work in a nutshell



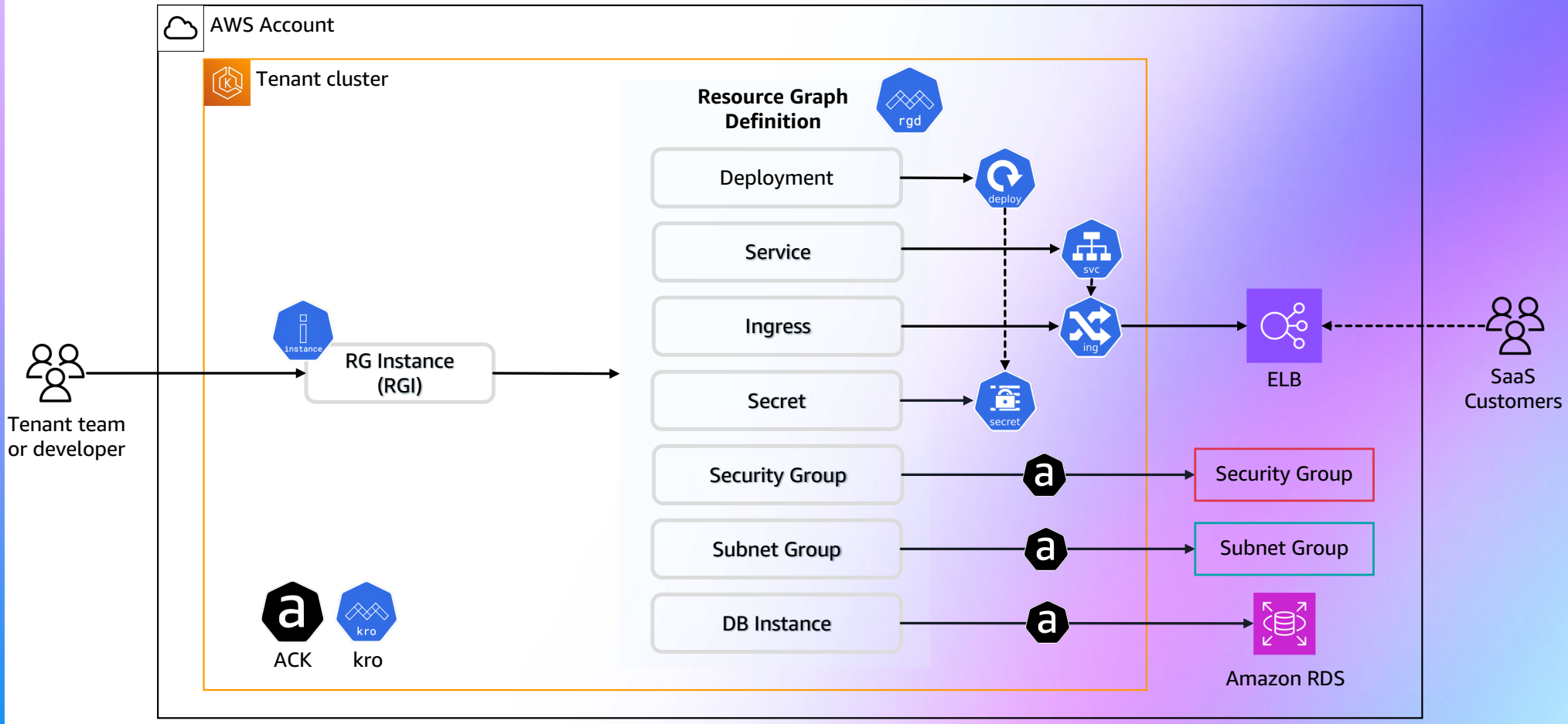
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How ACK and KRO work in a nutshell



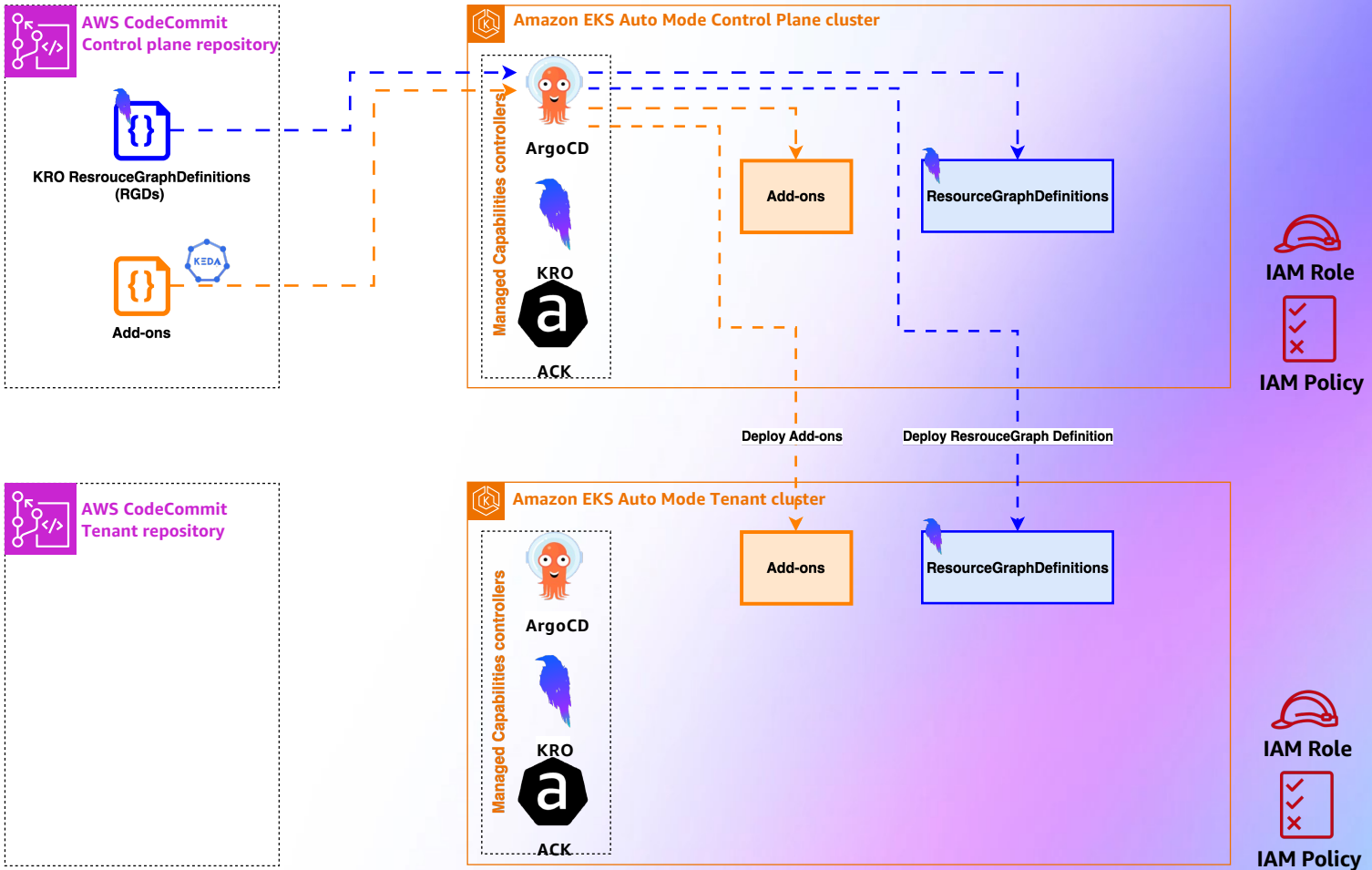
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How ACK and KRO work in a nutshell

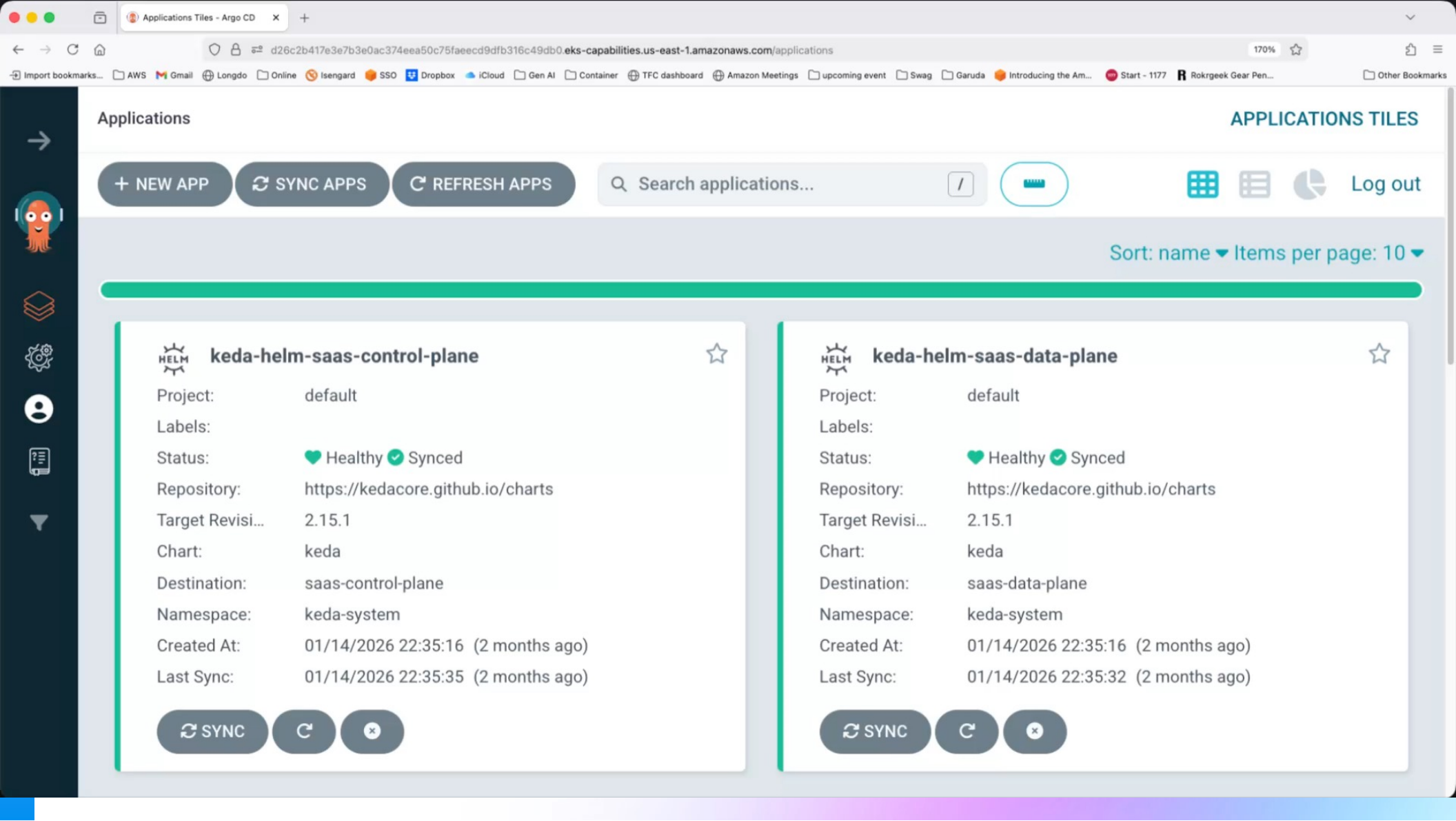


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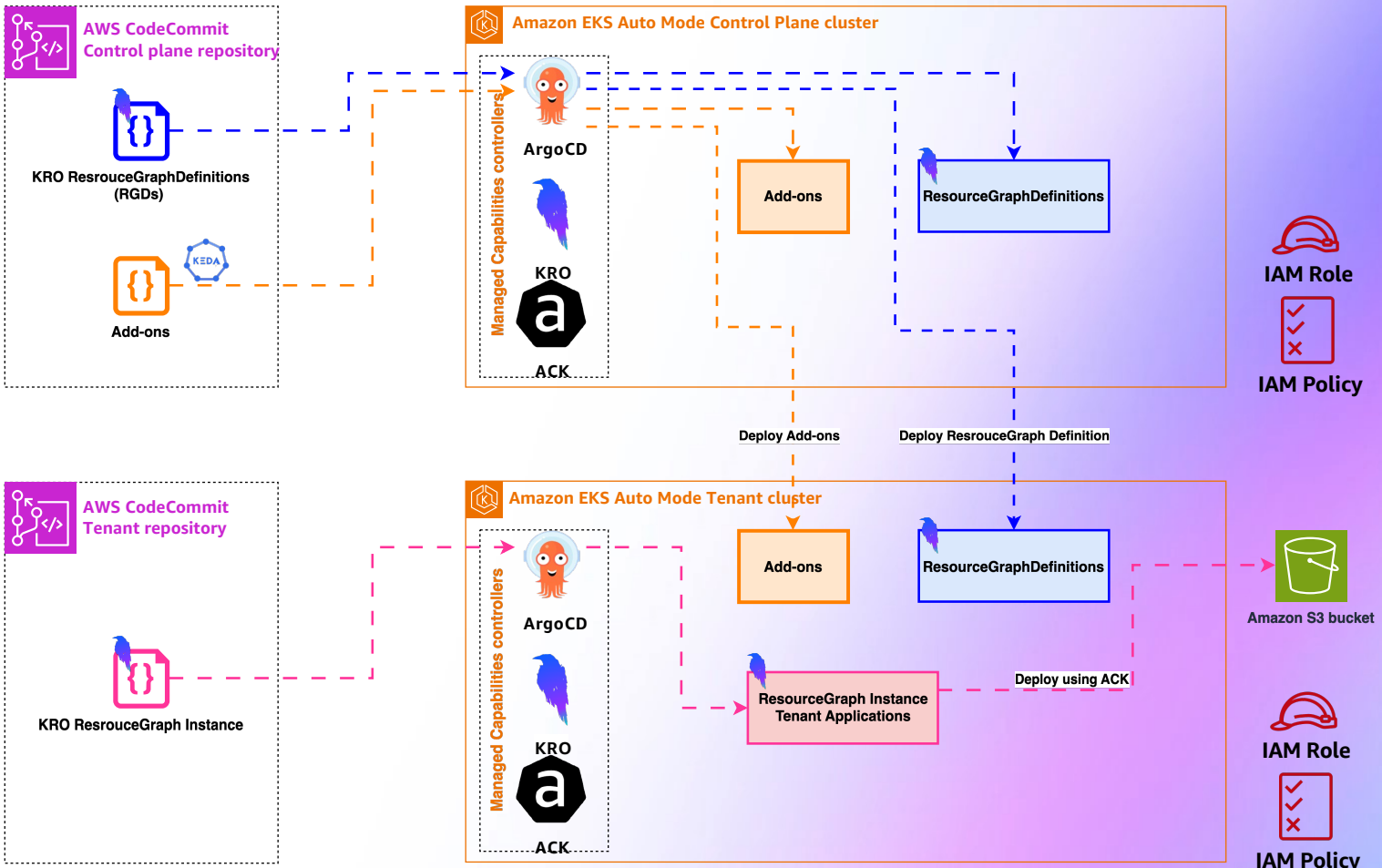
ResourceGraph definition deploy



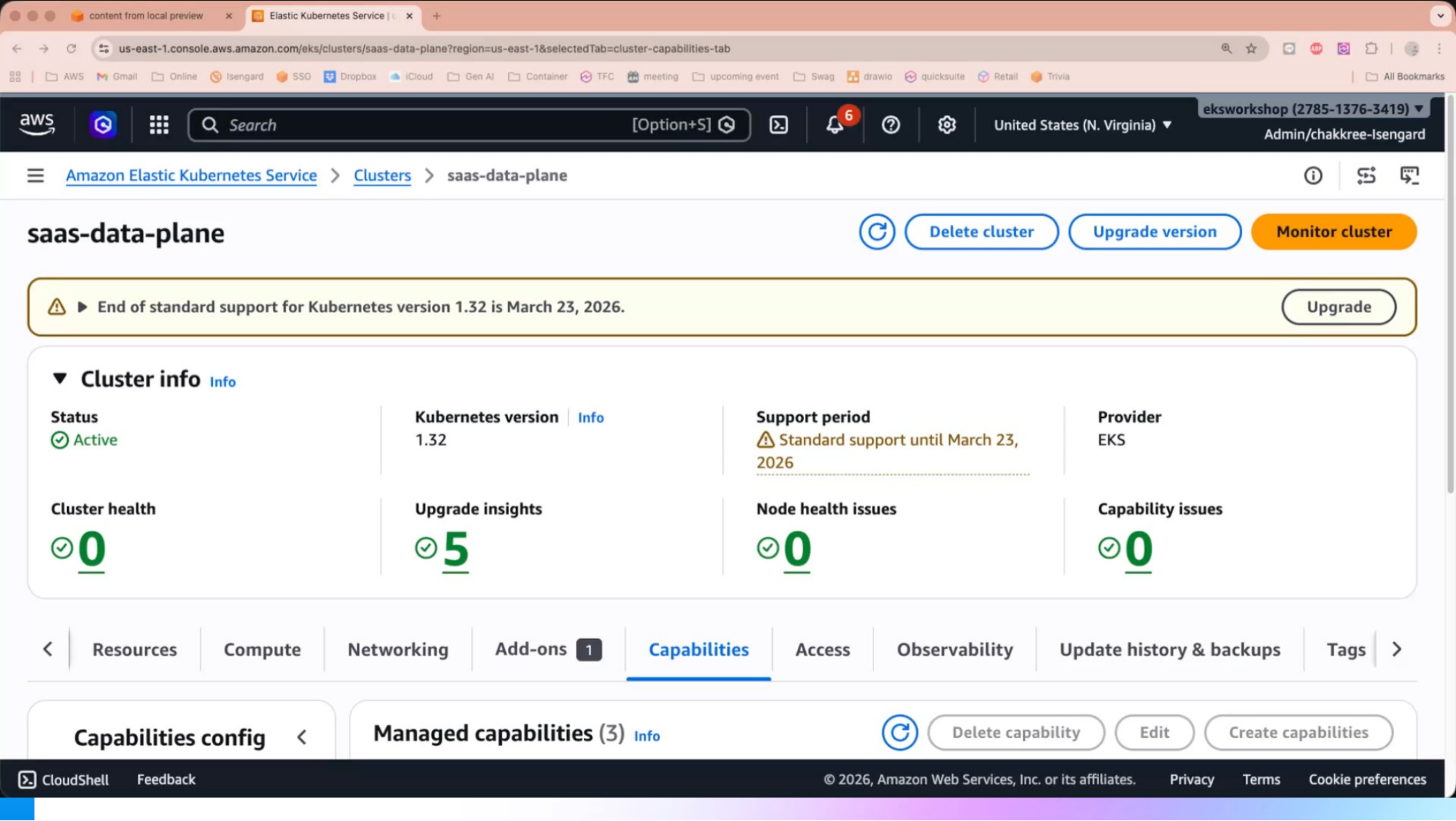
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Applications Tiles - Argo CD
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Applications
APPLICATIONS TILES
+ NEW APP
SYNC APPS
REFRESH APPS
Search applications...
Log out
Sort: name
Items per page: 10
keda-helm-saas-control-plane
Project: default
Labels:
Status: Healthy Synced
Repository: https://kedacore.github.io/charts
Target Revisi... 2.15.1
Chart: keda
Destination: saas-control-plane
Namespace: keda-system
Created At: 01/14/2026 22:35:16 (2 months ago)
Last Sync: 01/14/2026 22:35:35 (2 months ago)
SYNC
keda-helm-saas-data-plane
Project: default
Labels:
Status: Healthy Synced
Repository: https://kedacore.github.io/charts
Target Revisi... 2.15.1
Chart: keda
Destination: saas-data-plane
Namespace: keda-system
Created At: 01/14/2026 22:35:16 (2 months ago)
Last Sync: 01/14/2026 22:35:32 (2 months ago)
SYNC

Tenant deploys ResourceGraph Instance



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content from local preview
Elastic Kubernetes Service
us-east-1.console.aws.amazon.com/eks/clusters/saas-data-plane?region=us-east-1&selectedTab=cluster-capabilities-tab
Search [Option+S]
United States (N. Virginia)
eksworkshop (2785-1376-3419)
Admin/chakkree-Isengard
Amazon Elastic Kubernetes Service > Clusters > saas-data-plane
saas-data-plane
Delete cluster
Upgrade version
Monitor cluster
End of standard support for Kubernetes version 1.32 is March 23, 2026.
Upgrade
Cluster info Info
Status
Active
Kubernetes version Info
1.32
Support period
Standard support until March 23, 2026
Provider
EKS
Cluster health
0
Upgrade insights
5
Node health issues
0
Capability issues
0
Resources
Compute
Networking
Add-ons 1
Capabilities
Access
Observability
Update history & backups
Tags
Capabilities config
Managed capabilities (3) Info
Delete capability
Edit
Create capabilities
CloudShell
Feedback
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Cost Overview | **Tenant Service Costs**

Controls

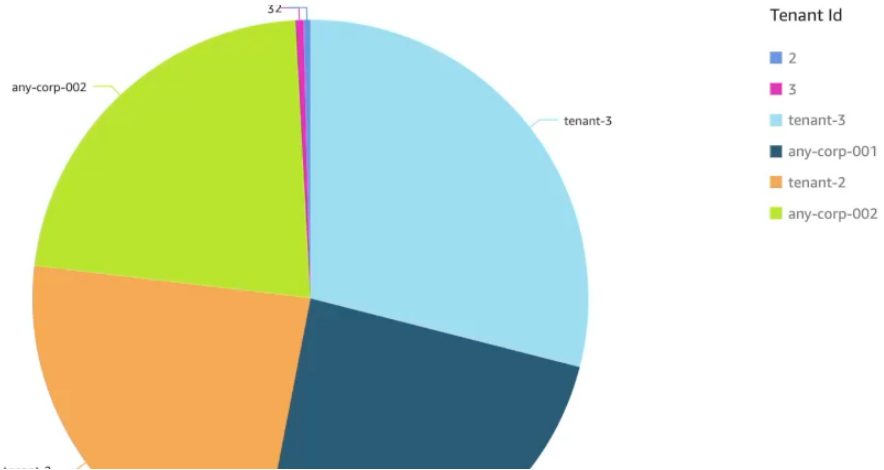
Tenant: All

Service: All

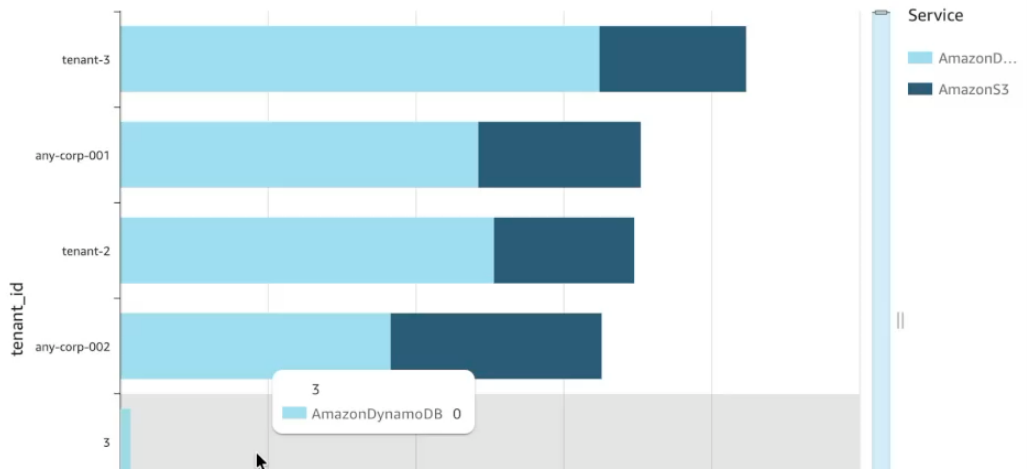
Total Tenant Service Cost

0.01

Cost by Tenant



Cost by Service per Tenant



Summary — Challenges Solved

- ✓ Platform Setup Complexity → EKS Auto Mode + EKS Capabilities
- ✓ Operational Drift → Control Plane Architecture + EKS Capabilities
- ✓ Slow Tenant Onboarding → Tenant Architecture + EKS Capabilities
- ✓ Unknown Costs → Split Cost Allocation + Amazon Quick Suite

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Thank you!

Chakkree Tipsupa

Solutions Architect
AWS

Rodrigo Bersa

Sr. Specialist SA, Containers
AWS

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Demo stations: Check out the rest of the AWS booth #700 with 4 demo stations and a Sketch Lab. Connect with AWS experts and ask your hardest questions!

Hands-on workshops: Register for online Kubernetes workshops with AWS subject matter experts covering AI/ML, platform engineering and much more...



Public roadmap: Stay up to date with what we’re working on, give us feedback and propose ideas to influence EKS roadmap

http://github.com/aws/containers-roadmap



Best Practices Guide: Master your Kubernetes operations and run production workloads with greater security, reliability, and efficiency

https://docs.aws.amazon.com/eks/latest/best-practices/introduction.html

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